

CHENJIA NI

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Education

University of Pennsylvania

Aug 2025 – May 2029

B.S. in Computer Science & Physics | GPA: 4.0

Philadelphia, PA

- Relevant Coursework: Stochastic Probability, AI Lab (Data, Systems & Decisions), Linear Algebra, PDEs
- Incoming **Teaching Assistant, CIS 1200** (Programming Languages & Techniques I) – Fall 2026
- Honors: Putnam Math Competition (Score: 10, top ~25%), USAPhO Bronze (2x), AIME Qualifier (5x), USACO Silver

Technical Skills

Languages: Python, C++, Java, R, JavaScript, HTML/CSS, $\text{\LaTeX}$

ML & Scientific Computing: PyTorch, scikit-learn, NumPy, Pandas, Brevitas, ONNX/QONNX, hls4ml, QuTiP, Qiskit

Web & Tools: Express.js, REST APIs, WebSockets, OpenAI API, Git, FPGA, Vivado/Vitis HLS

Experience

University of Pennsylvania, Rankin Lab (CERN)

Apr 2026 – Present

Undergraduate Researcher (Fast Machine Learning / FPGA)

Philadelphia, PA

- Implementing the **nano-PELICAN** Lorentz-equivariant network on **FPGAs** via **hls4ml** for low-latency jet-tagging inference in CERN's **CMS** trigger.
- Benchmarking **latency, throughput, and resource utilization** (LUTs, DSPs, BRAM) across quantization configurations to meet sub-microsecond trigger timing.

University of Pennsylvania, Zhou Lab

Nov 2025 – Apr 2026

Undergraduate Researcher (Bioinformatics)

Philadelphia, PA

- Engineered a **Random Forest** cancer-detection pipeline in **R**, comparing **Euclidean vs. Spearman distance metrics** to improve cytometric feature-space representation.
- Implemented **dithering-based data augmentation** to expand training sets, achieving **3-5% gains in classification accuracy** on validation.

Colgate University Physics Department

Jan 2024 – Jun 2024

Student Research Intern (Quantum Information)

Hamilton, NY

- Built **Hamiltonian** and **Lindblad** models for a trapped-ion qubit chain; simulated decoherence with **QuTiP** in **Python** to evaluate T_2 vs. ion spacing and noise.
- Simplified experimental simulations, reducing trials from **hours to seconds**, replicating real ion-trap coherence behavior.

Projects

PELICAN-nano (QAT) | *Source Code*

PyTorch | Brevitas | ONNX / hls4ml

- Added **Brevitas QAT** to the nano-PELICAN Lorentz-equivariant top-tagger, holding **AUC $\approx$ 0.96 down to 6-bit** vs. as low as 0.40 under post-training quantization.
- Built an **ONNX/QONNX export** pipeline feeding the **hls4ml** FPGA flow, dumping integer weights and scales for the firmware loader.

ClassMate | *Source Code*

HTML | Express.js | OpenAI Whisper

- Engineered a full-stack web app with real-time group formation, chat, and file-sharing using **socket-based communication** and **RESTful APIs**.
- Integrated an asynchronous **OpenAI Whisper API** pipeline for lecture transcription and summarization into the app data model.

Leadership & Campus

Wharton Investment & Trading Group — Quantitative Investment Strategies

Sep 2025 – Present

Member & Research Analyst

Philadelphia, PA

- Programmed a **Python trading bot** for futures markets, implementing signal generation, position sizing, and PnL.
- Built a **DCF** and **comparables model** for an equity pitch on **Coursera (NYSE: COUR)**, analyzing edtech and AI-integration trends.